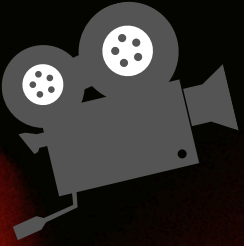


NETFLIX

Exploratory Data Analysis



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About the Netflix

Netflix is one of the world's leading streaming entertainment platforms, offering a vast library of movies, TV shows, documentaries, and original productions across multiple genres and languages. Founded in 1997 as a DVD-rental-by-mail service, Netflix transformed its business model in 2007 by launching online streaming—an innovation that revolutionized how audiences consume content. Today, Netflix operates in more than 190 countries and serves millions of subscribers who access content anytime, anywhere, through internet-connected devices such as smartphones, smart TVs, laptops, and gaming consoles.

Over the years, Netflix has invested heavily in content creation, producing globally acclaimed originals including *Stranger Things*, *Money Heist*, *The Crown*, and *Narcos*. Its data-driven approach allows the platform to understand viewer behavior, personalize recommendations, and enhance user engagement. By analyzing user preferences and viewing patterns, Netflix continuously improves its content offerings and user experience.

Netflix remains a dominant force in the streaming industry. As the digital entertainment landscape evolves, Netflix continues to shape the future of on-demand content worldwide.

Objective

Purpose of the Dataset:

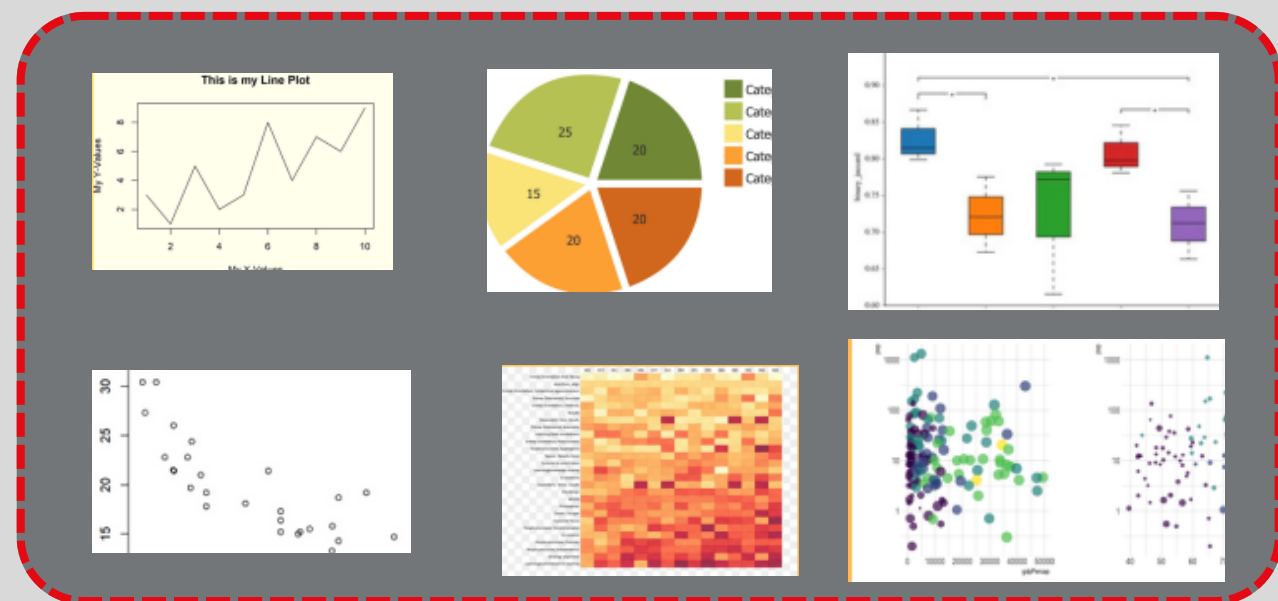
The goal of the Netflix EDA project is to conduct a comprehensive exploration and analysis of Netflix's content dataset. This includes understanding the data structure, ensuring data integrity by handling missing values and duplicates, deriving descriptive statistics, and visualizing content distribution across genres and release years.

Additionally, the project aims to identify temporal trends, analyze content attributes like ratings and duration, and assess audience engagement metrics. By synthesizing these insights, the project aims to draw meaningful conclusions and provide actionable recommendations to enhance Netflix's content offerings and user experience.

| TOOLS



| GRAPHS



Data Overview

- Loading and preparing loan application data for analysis in google Colab notebook.

```
[ ] from google.colab import drive  
drive.mount('/content/drive')
```

Mounted at /content/drive

- In this analysis, we import **Pandas** for data manipulation, **Numpy** for numerical operations, **Matplotlib** for creating visualization and **Seaborn** and **plotly** for enhanced statistical graphic.

```
import numpy as np  
import pandas as pd  
import matplotlib.pyplot as plt  
import seaborn as sns  
import plotly.express as px
```

- We import the **Google drive** module to mount Google `Drive`, enabling access to files and dataset in the for our analysis.

```
a = "/content/drive/MyDrive/Netflix_eda/netflix_titles.csv"
```

```
df = pd.read_csv(a)
```

Data Overview

- We use **pandas** to Read the CSV file containing big vehicle insurance from Google Drive, loading it into a DATAFRAME name 'df' for analysis.

overview of dataset

df												
show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description	
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Oamala, Khosi Ngema, Gail Mablane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Lederq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabl...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train l...
...	...	...	...	...	...	...	...	...	...	...	...	...
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	July 1, 2019	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	November 1, 2019	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	January 11, 2020	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	March 2, 2019	2015	TV-14	111 min	Dramas, International Movies, Music & Musicals	A scrappy but poor boy worms his way into a ty...

8807 rows x 12 columns

The dataset comprises of **8807** rows and **12** columns

Understanding the columns

- First Few rows

```
df.head()
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	genres	description	year_added	month_added
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Unknown	United States	2021-09-25	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, film...	2021	9
1	s2	TV Show	Blood & Water	Unknown	Ama Oamala, Khosi Ngema, Gail Mablane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town L...	2021	9
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabl...	Unknown	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...	2021	9
3	s4	TV Show	Jailbirds New Orleans	Unknown	Unknown	Unknown	2021-09-24	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...	2021	9
4	s5	TV Show	Kota Factory	Unknown	Mayur More, Jilendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train L...	2021	9

Description of columns

- **show_id**: Each title on Netflix has a unique identifier (e.g., s1). This helps in tracking and referencing individual movies or TV shows.
- **type**: Indicates whether the title is a Movie or a TV Show. This helps categorize content based on format.
- **title**: The official name of the movie or TV show (e.g., Money Heist). It is used to identify the content.
- **director**: This column lists the director(s) associated with the title (e.g., Christopher Nolan). It may be empty for some titles. It helps analyze content based on creators.

Understanding the columns

- **cast:** Contains the names of actors featured in the title (e.g., Tom Hanks, Leonardo DiCaprio). Useful for understanding actor popularity and collaborations.
- **country:** Shows the country or countries where the title was produced (e.g., United States, India). Helps analyze regional content distribution.
- **date_added:** The date on which Netflix added the title to its platform (e.g., September 9, 2019). Useful for studying content release timelines.
- **release_year:** The year the movie or TV show was originally released (e.g., 2018). Helps analyze trends in content production over the years.
- **rating:** Indicates the audience maturity rating (e.g., TV-MA, PG-13, G). This shows the target audience age group for each title.

Understanding the columns

- **duration:**
 1. For movies: Shows runtime in minutes (e.g., 120 min).
 2. For TV shows: Shows number of seasons (e.g., 2 Seasons). Helps compare content length.
- **listed_in**: Lists the genres or categories assigned to each title (e.g., Drama, Comedy, Action). Helps identify popular content themes.
- **description**: A short summary of the movie or TV show's storyline or theme. Helps understand the plot and classify content.

Data Exploration

```
df.shape
```

```
(8807, 12)
```

The dataset comprises of **8807** rows and **12** columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   show_id         8807 non-null   object
1   type            8807 non-null   object
2   title           8807 non-null   object
3   director        6173 non-null   object
4   cast            7982 non-null   object
5   country         7976 non-null   object
6   date_added      8797 non-null   object
7   release_year    8807 non-null   int64
8   rating          8803 non-null   object
9   duration        8804 non-null   object
10  listed_in       8807 non-null   object
11  description     8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

Explanation of Dataset Information

The Netflix dataset consists of 8,807 records and 12 columns, capturing essential metadata about movies and TV shows available on the platform. The dataset includes 11 categorical features and 1 numerical feature. These variables describe content attributes such as title, director, cast, duration, genres, country of origin, and maturity rating. A few columns, including director, cast, and country, contain missing values.

Data Cleaning

- Checking for missing Values

By values

```
df.isnull().sum()
```

	0
show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
listed_in	0
description	0

By Percentage

```
df.isnull().sum()/df.shape[0]*100
```

	0
show_id	0.000000
type	0.000000
title	0.000000
director	29.908028
cast	9.367549
country	9.435676
date_added	0.113546
release_year	0.000000
rating	0.045418
duration	0.034064
listed_in	0.000000
description	0.000000

summary of missing values:

The dataset is mostly complete, with missing values concentrated in a few metadata fields. The director column shows the highest number of missing entries (2,634), then cast (825) and country (831), indicating incomplete creative and geographic information for some titles. Minor missing values are present in date_added (10), rating (4), and duration (3).

Data Exploration

- **handling missing Values**

```
df['director']=df['director'].fillna("Unknown")
df['cast']=df['cast'].fillna("Unknown")
df['country']=df['country'].fillna("Unknown")
df["duration"].fillna(df["duration"].mode()[0],inplace=True)
```

filling 'director', 'cast', 'country', 'duration' with unknown.

```
df.dropna(subset=['rating','duration'], inplace=True)
```

Dropping missing values of rating and duration .

- **Remove duplicate entries if any.**

```
df.duplicated().sum()
np.int64(0)
```

- **Correct any inconsistencies or errors in the data.**

```
df.rename(columns={'listed_in':'genres'},inplace=True)

df["year_added"] = pd.to_numeric(df["year_added"],
errors='coerce').fillna(0).astype(int)
df["month_added"] = pd.to_numeric(df["month_added"],
errors='coerce').fillna(0).astype(int)
```

renaming listed_in column to genres for better understanding and converting columns year_added and month_added to int for analysis purpose.

Data Exploration

- Summarize basic statistics for the numeric columns.

```
df.describe()
```

	date_added	release_year	year_added	month_added
count	8705	8803.000000	8803.000000	8803.000000
mean	2019-05-23 10:08:15.439402496	2014.179825	1996.413382	6.580257
min	2008-01-01 00:00:00	1925.000000	0.000000	0.000000
25%	2018-04-20 00:00:00	2013.000000	2018.000000	4.000000
50%	2019-07-12 00:00:00	2017.000000	2019.000000	7.000000
75%	2020-08-26 00:00:00	2019.000000	2020.000000	10.000000
max	2021-09-25 00:00:00	2021.000000	2021.000000	12.000000
std	NaN	8.821247	211.843699	3.482192

Summary of Descriptive Statistics

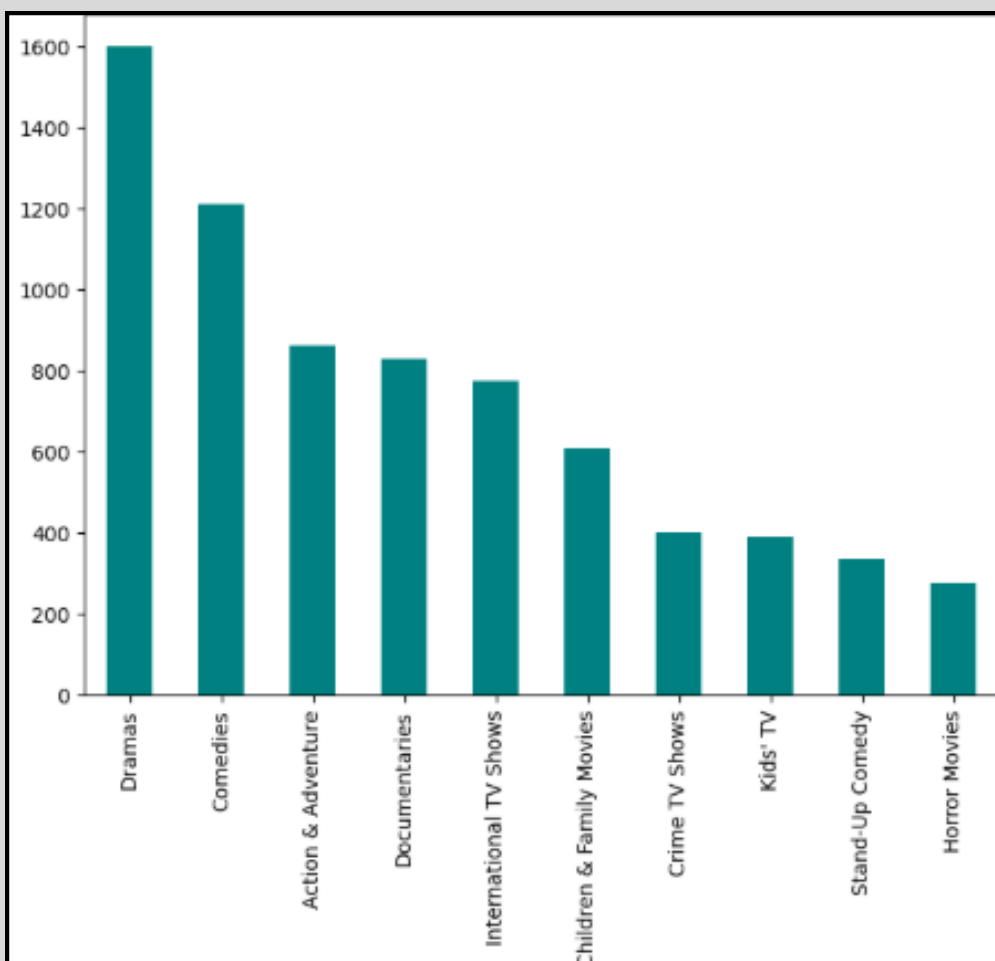
The dataset shows that most titles on Netflix were added between 2018 and 2021, with the median date added being July 2019. The release years range widely from 1925 to 2021, but the median release year is 2017, indicating that Netflix's catalog is largely dominated by modern content.

Overall, the descriptive statistics highlight that Netflix's library is primarily composed of recent releases, with significant expansion in the 2018–2021 period.

Data Visualization

- Create visualizations to represent the distribution of content over different genres.

```
top_genres = df['genres'].str.split(',').str.get(0)
plt.figure(figsize=(10,6))
top_genres.value_counts().head(10).plot(kind="bar",color="teal")
plt.title("Distribution of content over different genres",fontsize=20)
plt.show()
```



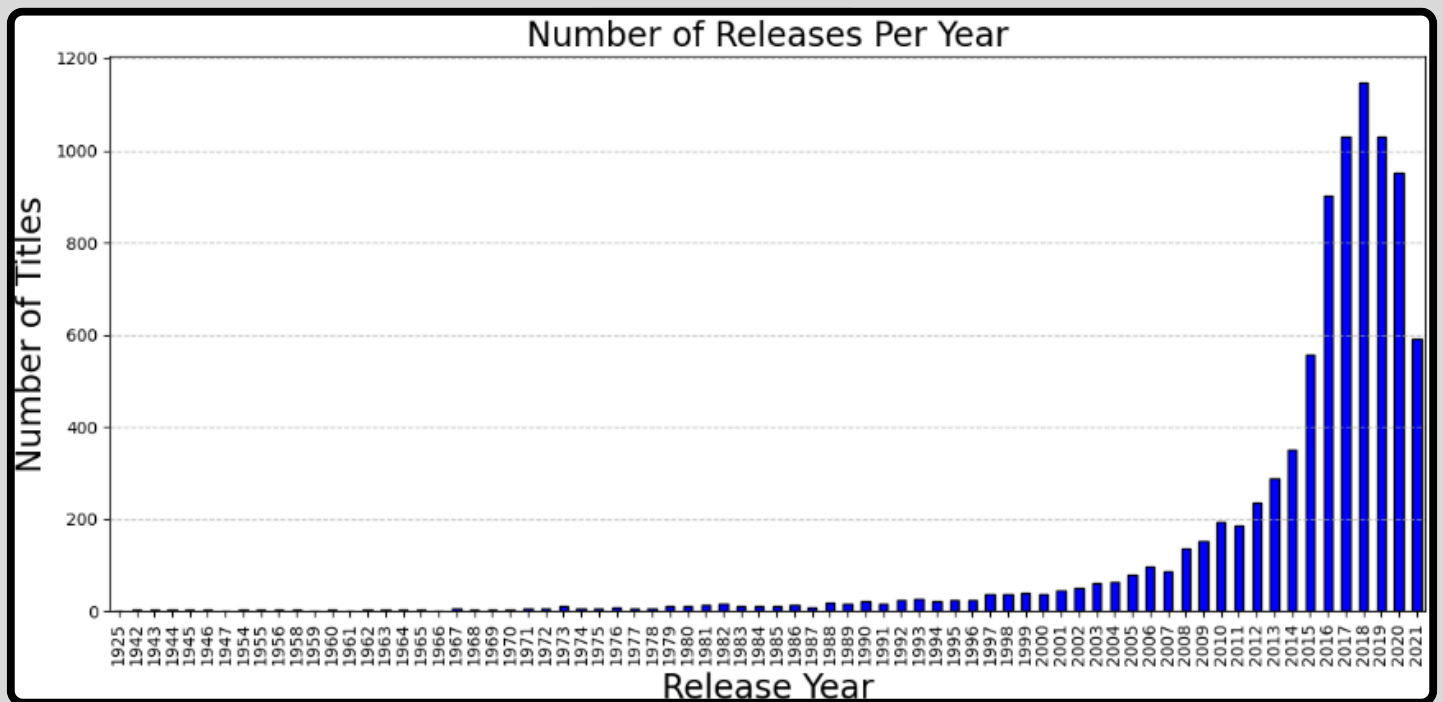
Insights:

The analysis of the top genres shows that a few categories dominate Netflix's library. Genres such as Dramas, Comedies, and Documentaries appear most frequently, indicating a strong user demand and a content strategy focused on these popular categories.

Data Visualization

- Visualize the distribution of content across release years.

```
plt.figure(figsize=(12, 6))
df['release_year'].value_counts().sort_index().plot(kind='bar',
color='blue',edgecolor='black')
plt.title('Number of Releases Per Year',fontsize=20)
plt.xlabel('Release Year',fontsize=20)
plt.ylabel('Number of Titles',fontsize=20)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```



Insights:

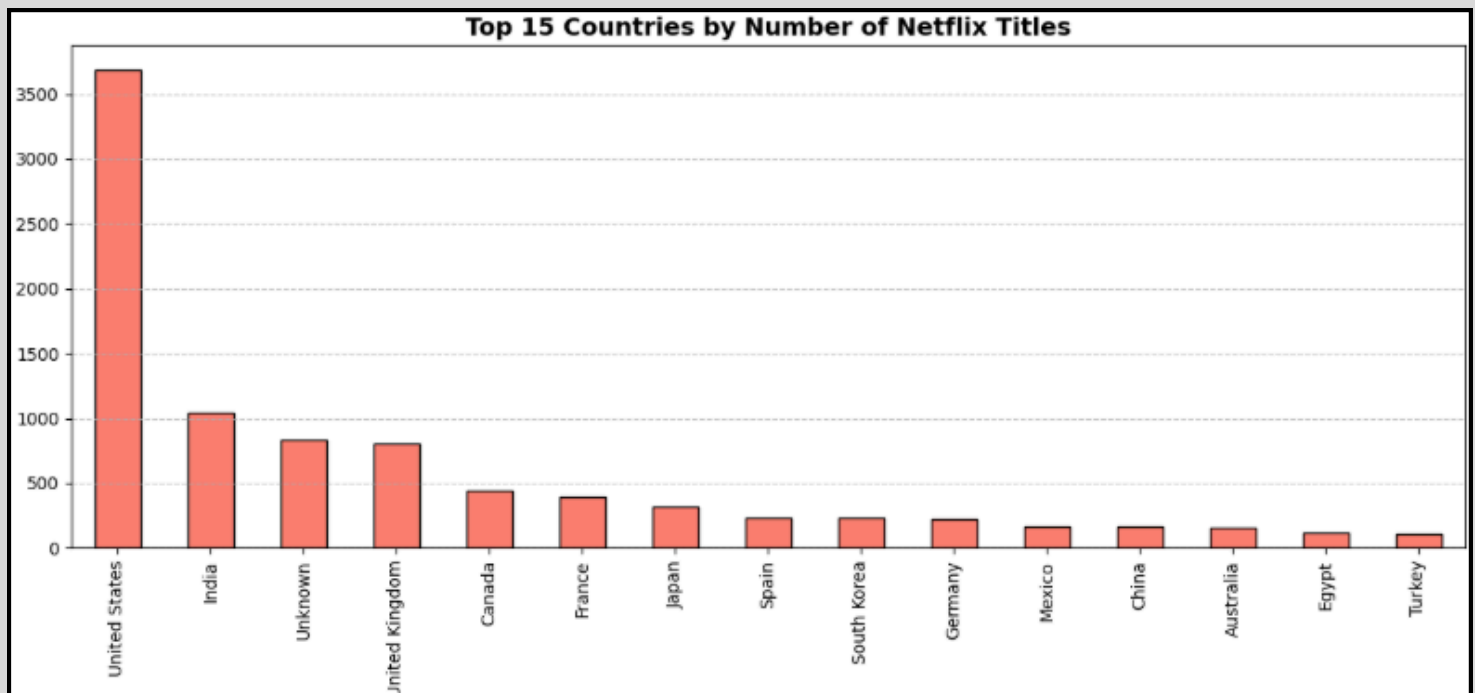
The number of titles rises sharply after 2000, peaking around 2017–2020 — this reflects Netflix’s rapid global content expansion during that period.

A steep decline appears after 2020, likely due to production delays caused by the COVID-19 pandemic. Older movies (before 1990) are much fewer, indicating Netflix’s primary focus on modern and recent releases.

Data Visualization

- Explore the geographical distribution of content.

```
country_counts = df['country'].dropna().str.split(',').explode().str.strip().value_counts().head(15)
plt.figure(figsize=(12,6))
country_counts.plot(kind='bar', color='salmon', edgecolor='black')
plt.title('Top 15 Countries by Number of Netflix Titles', fontsize=14, fontweight='bold')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```

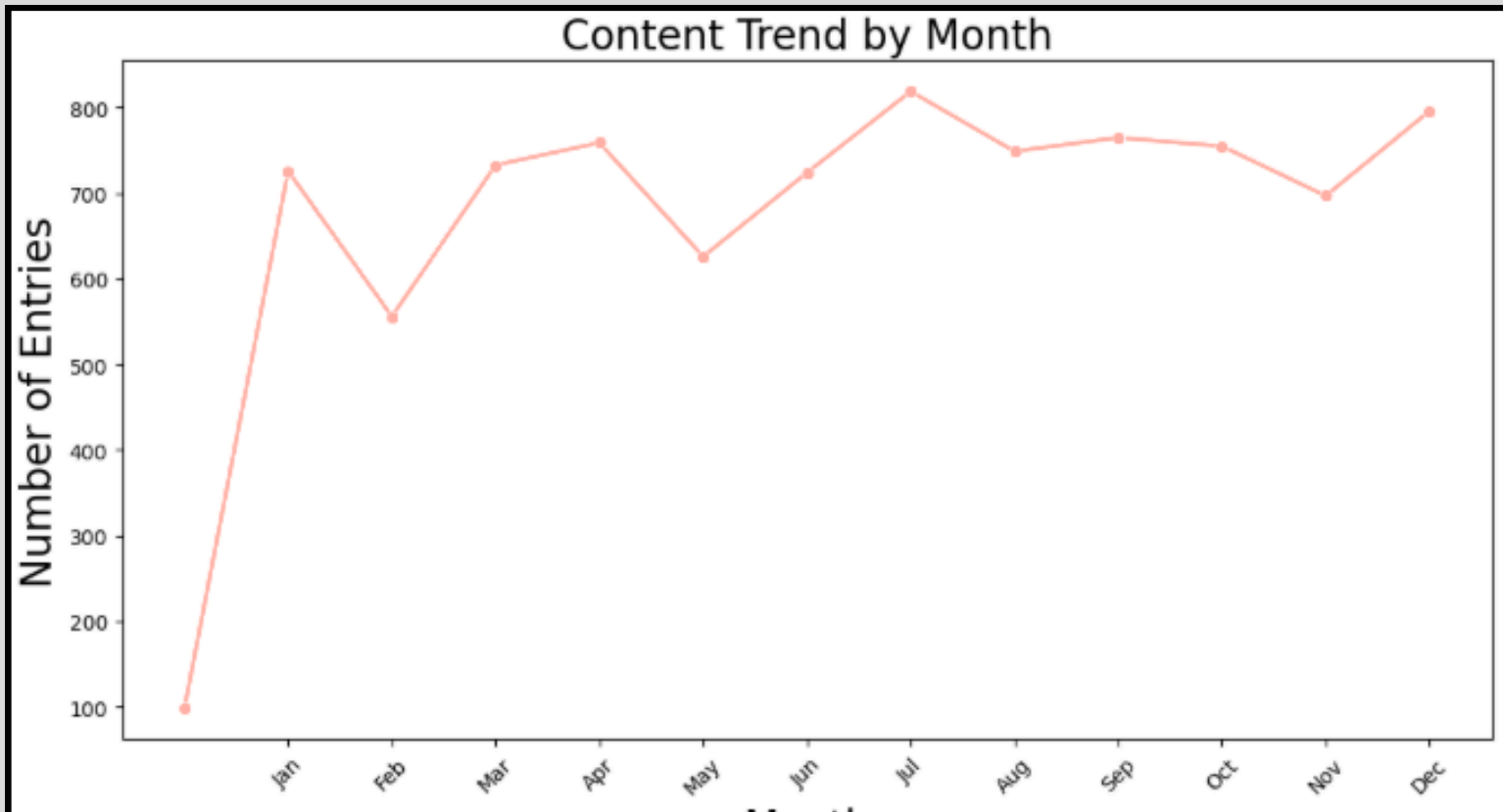


Insights:

- United States dominates Netflix's catalog, followed by India, United Kingdom, and Japan.
- This suggests Netflix has a strong focus on content from North America and Asia.
- The diversity in countries like South Korea, France, and Canada shows Netflix's growing global production strategy.

Data Visualization

- **Time Series Analysis:** If there's a temporal component, perform time series analysis to identify trends and patterns over time.



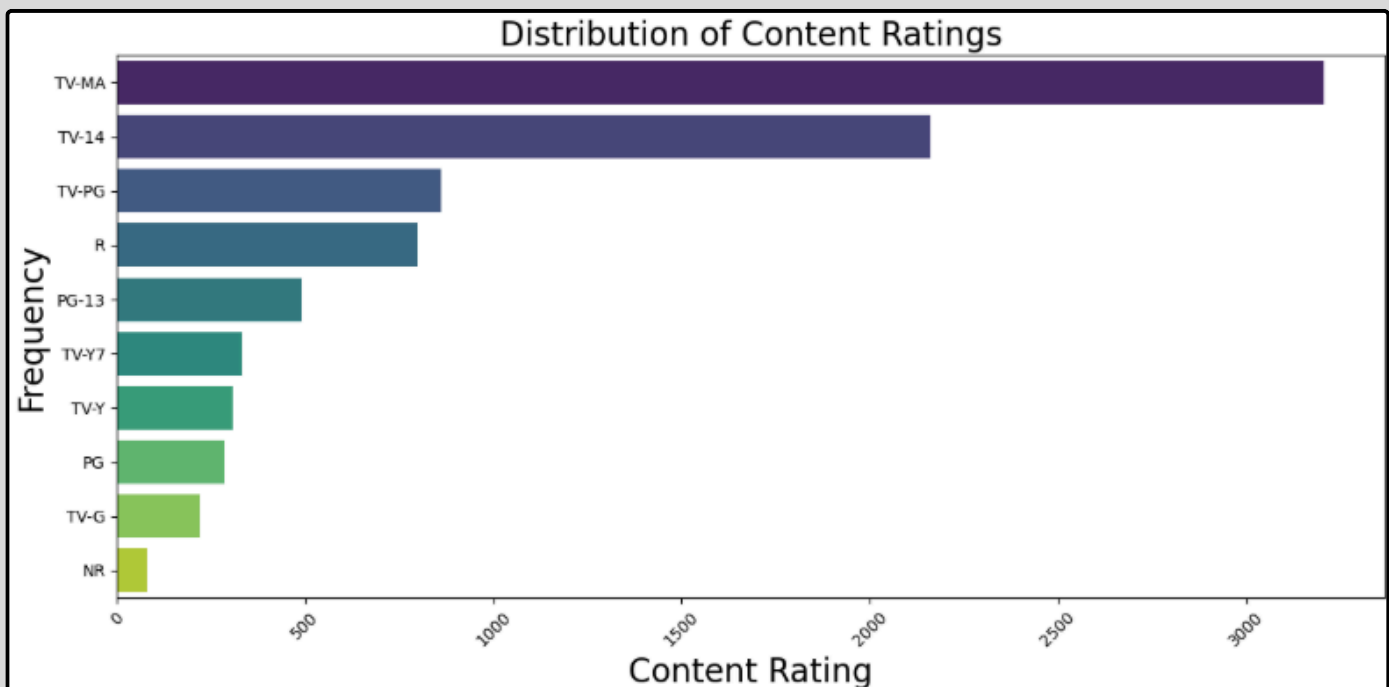
Insights:

- July sees the highest number of content additions, suggesting a mid-year content push.
- Other peak months include December and April, aligning with holiday seasons and Q4 strategy.
- January shows a sharp dip, possibly due to carry-over effects from year-end or data gaps.
- Overall, content is consistently added throughout the year, with a slight preference for mid-to-late months.

Data Visualization

- Analyze the distribution of content ratings

```
plt.figure(figsize=(12, 6))
sns.countplot(data=df, y='rating',
order=df['rating'].value_counts().index[:10], palette='viridis')
plt.title('Distribution of Content Ratings',fontsize=20)
plt.xlabel('Content Rating',fontsize=20)
plt.ylabel('Frequency',fontsize=20)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

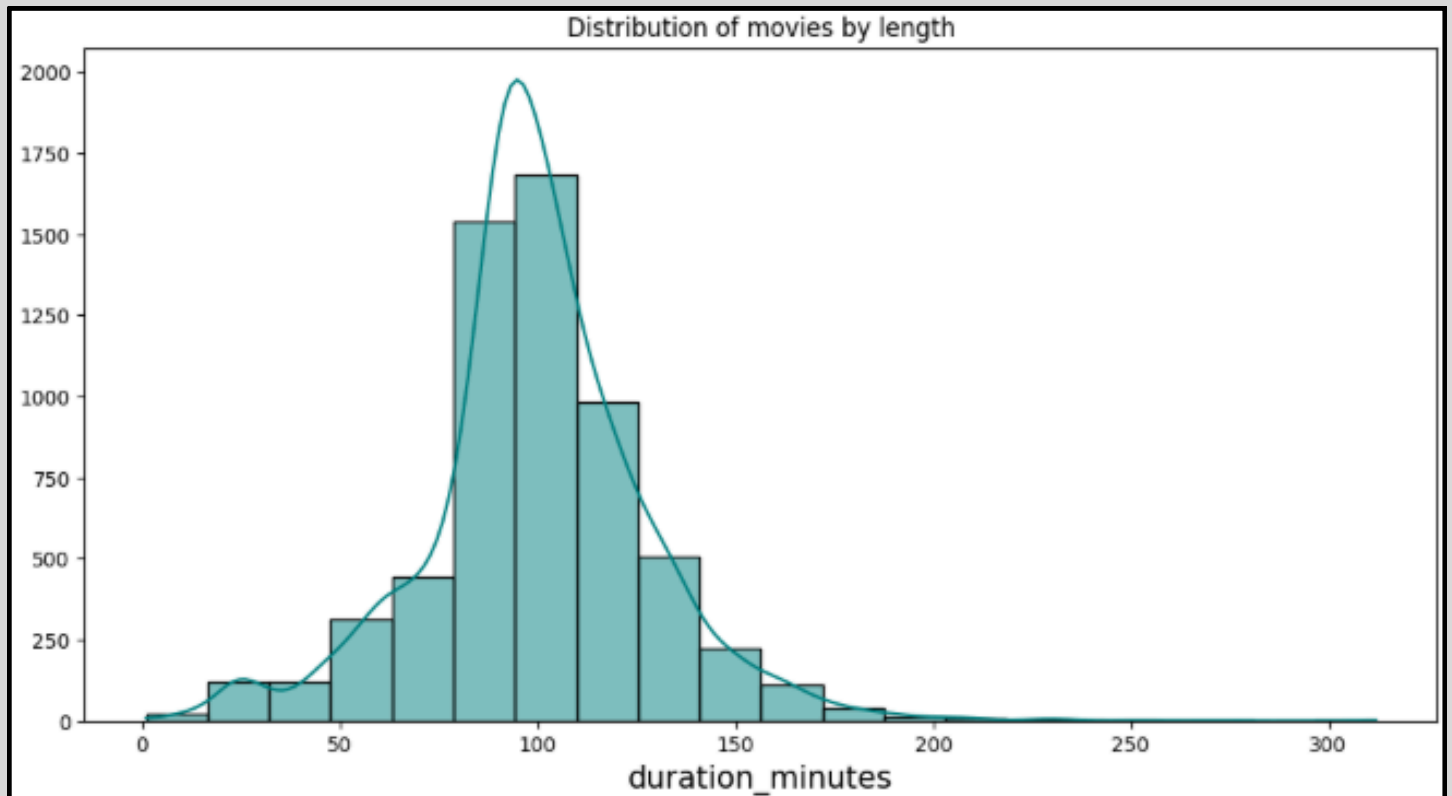


Insights:

- TV-MA is the most frequent rating, indicating a large portion of content is targeted at mature audiences.
- Other dominant ratings include TV-14, TV-PG, and R, showing a balanced mix of teen to adult content.
- Few titles are rated for children (like TV-Y and G), suggesting less focus on kids' programming.
- The variety in ratings reflects Netflix's strategy to cater to a wide audience, with a strong skew toward adult viewers.

Data Visualization

- Explore the length of movies or episodes and identify any trends.

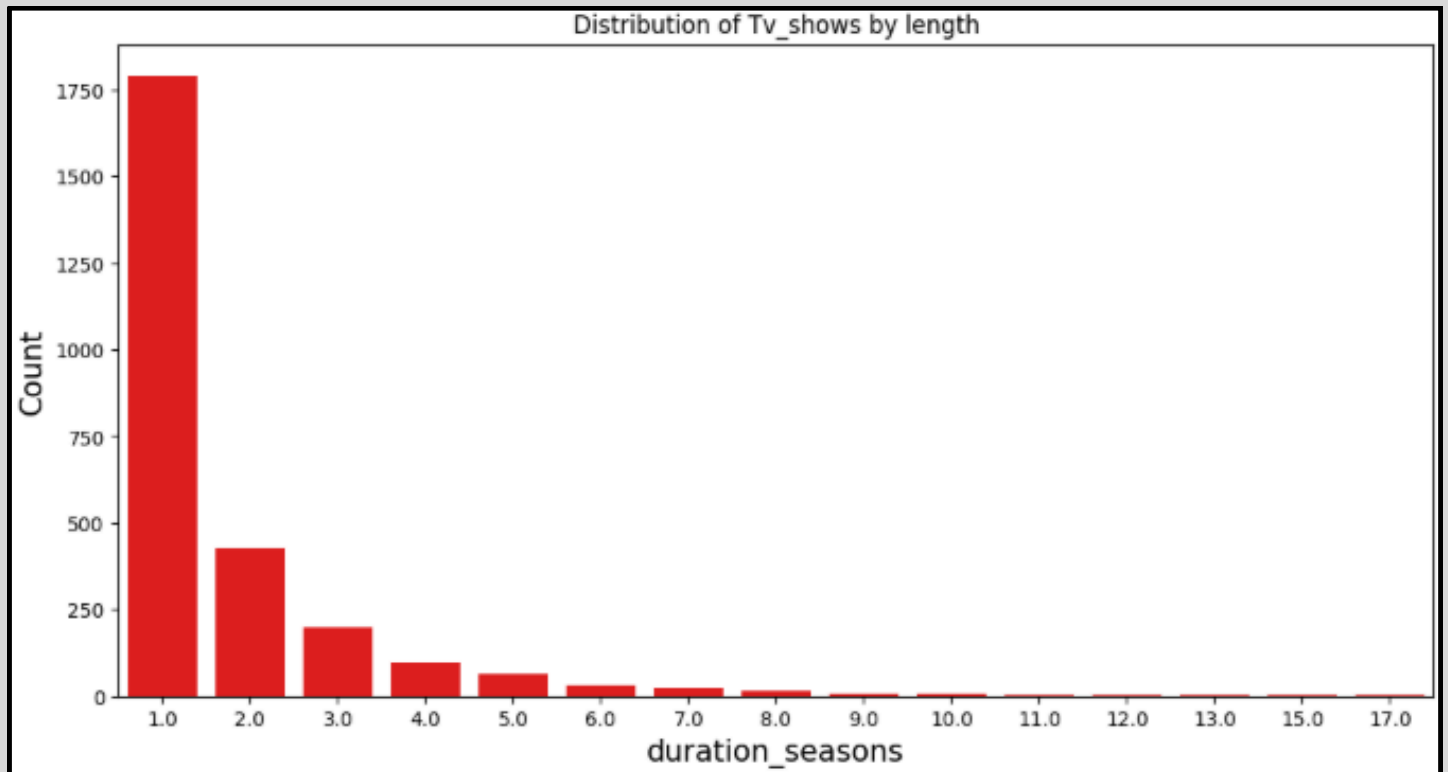


Insights:

- Most Netflix movies range between 80–120 minutes.
- There is a high concentration of movies around 90 minutes, indicating Netflix prefers shorter, easy-to-consume content.
- Very short movies (under 60 minutes) exist but are relatively uncommon; these are mostly documentaries or comedy specials.
- Very long movies (over 150 minutes) are rare and represent a small number of titles, often international films or special releases.

Overall, the distribution is right-skewed, meaning there are fewer extremely long movies compared to shorter ones.

Data Visualization



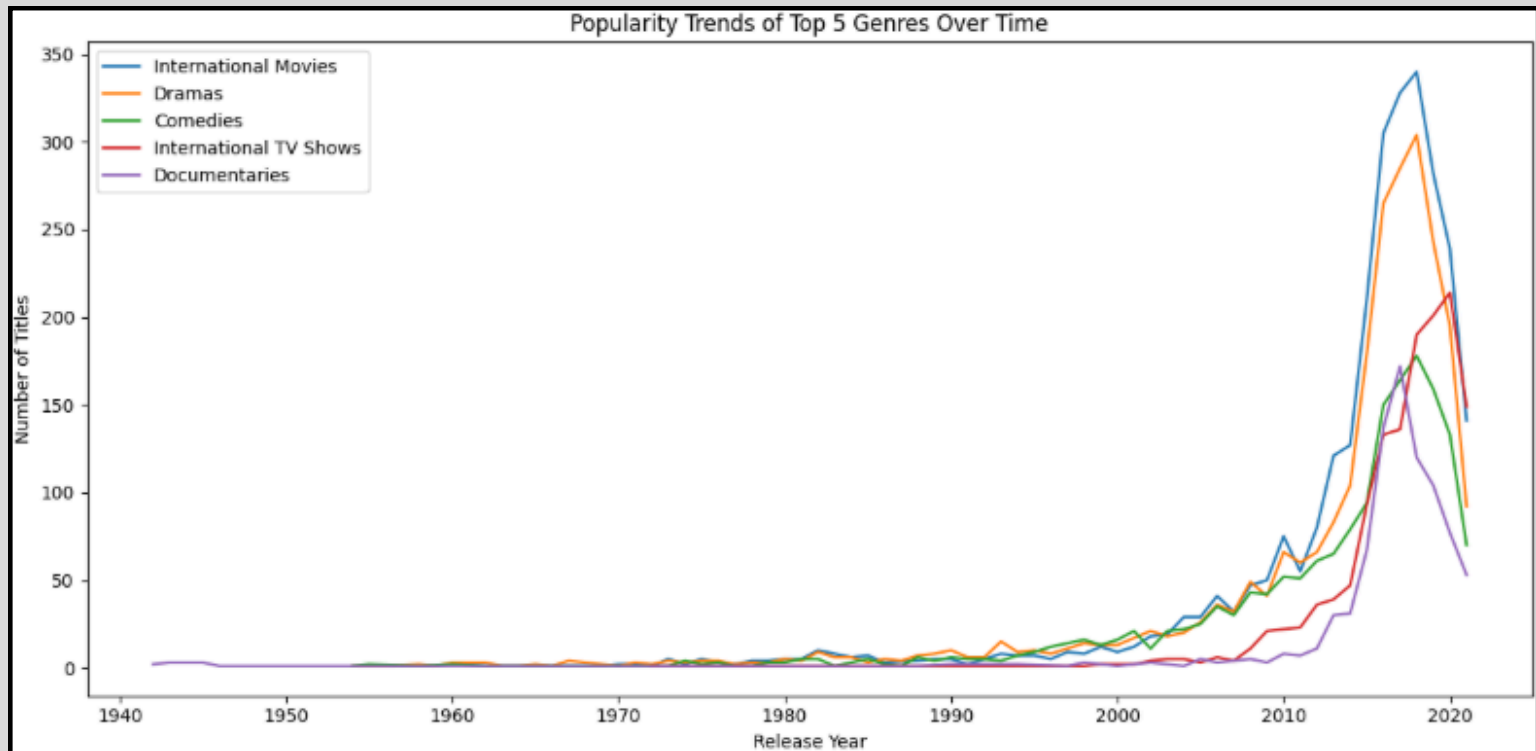
Insights:

- Most TV shows on Netflix have only 1 season. This suggests Netflix has a large volume of mini-series, limited series, and newly released shows.
- Shows with 2–3 seasons form the next biggest chunk, representing typical short-format streaming series.
- Shows with more than 5 seasons are rare, indicating Netflix has fewer long-running, traditional television series compared to networks like CBS or NBC.
- The distribution is heavily skewed toward fewer seasons, showing Netflix relies heavily on fresh titles and limited-run content rather than long multi-season shows.

Data Visualization

Genre Trends:

- Analyze trends in the popularity of different genres over time.



Insights:

1. Significant Increase in Genre Diversity Over Time

The number of titles across all major genres shows a clear upward trend, especially after 2015. Early years had relatively fewer releases, whereas later years reflect a broader and more diverse content catalog. This indicates Netflix's strategic expansion and higher production volume in recent years.

Data Visualization

2.Rapid Growth in International Content

International genres demonstrate a notable rise, particularly after 2018. This trend highlights Netflix's global expansion strategy and increasing investment in region-specific content to non-English-speaking markets.

3.Steady Rise in Documentary Content

Documentary titles exhibit consistent growth across the years. The steady upward trend suggests increased audience interest in factual, reality-based storytelling and Netflix's focus on producing documentary films and series.

4.Dominance of Drama-Based Genres

Drama remains one of the most consistently popular and frequently released genres throughout the observed period.

5.Increasing Popularity of Crime-Related Genres

Crime genres show a noticeable surge, particularly after 2016. This rise aligns with the growing popularity of crime thrillers and investigative series on streaming platforms.

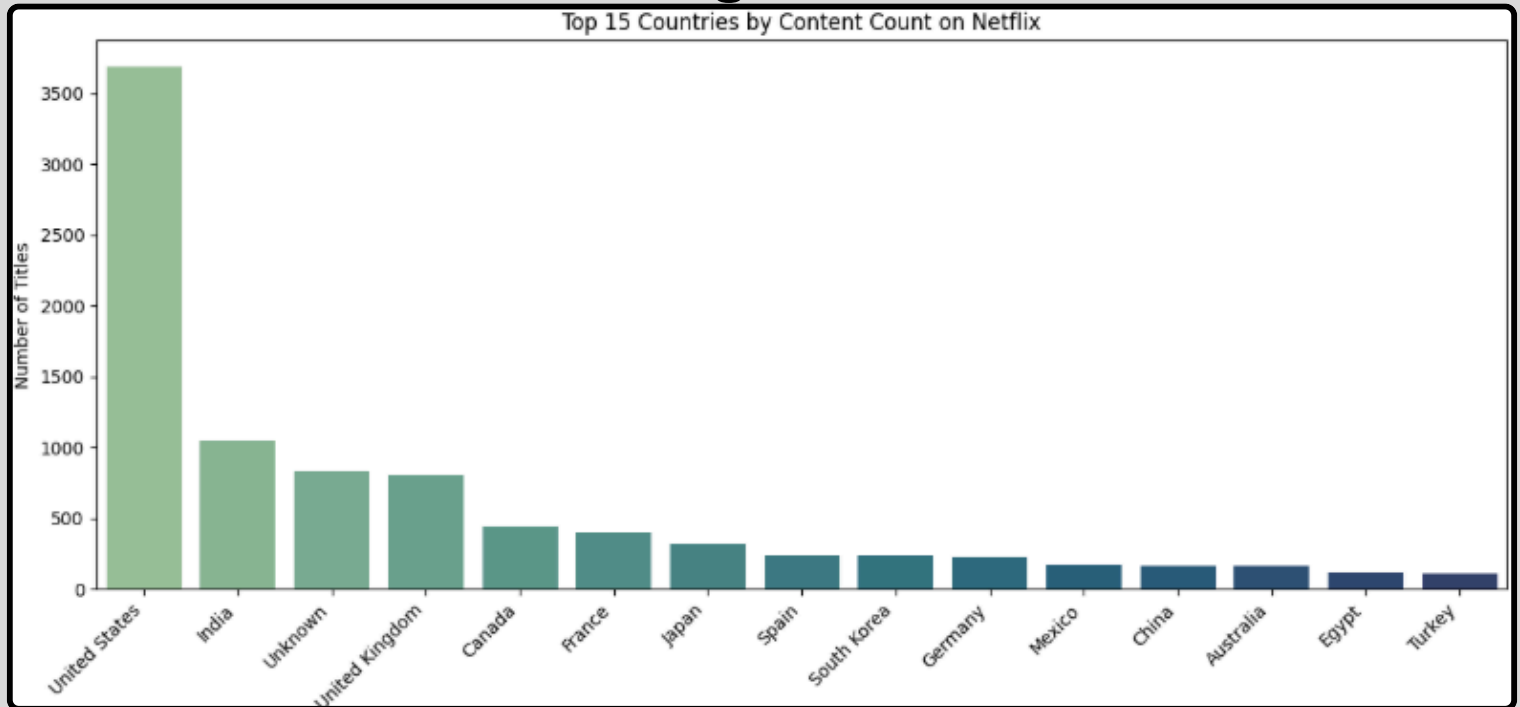
6.Sharp Increase Around 2020

Most genres exhibit a pronounced spike in the number of releases around 2020. This trend corresponds to the COVID-19 pandemic period, during which streaming consumption increased significantly, prompting Netflix to accelerate the release of new content across multiple genres.

Data Visualization

Geographical Analysis:

- Further explore the distribution of content across different countries and regions.



Insights:

- Most TV shows on Netflix have only 1 season. This suggests Netflix has a large volume of mini-series, limited series, and newly released shows.
- Shows with 2–3 seasons form the next biggest chunk, representing typical short-format streaming series.
- Shows with more than 5 seasons are rare, indicating Netflix has fewer long-running, traditional television series compared to networks like CBS or NBC.
- The distribution is heavily skewed toward fewer seasons, showing Netflix relies heavily on fresh titles and limited-run content rather than long multi-season shows.

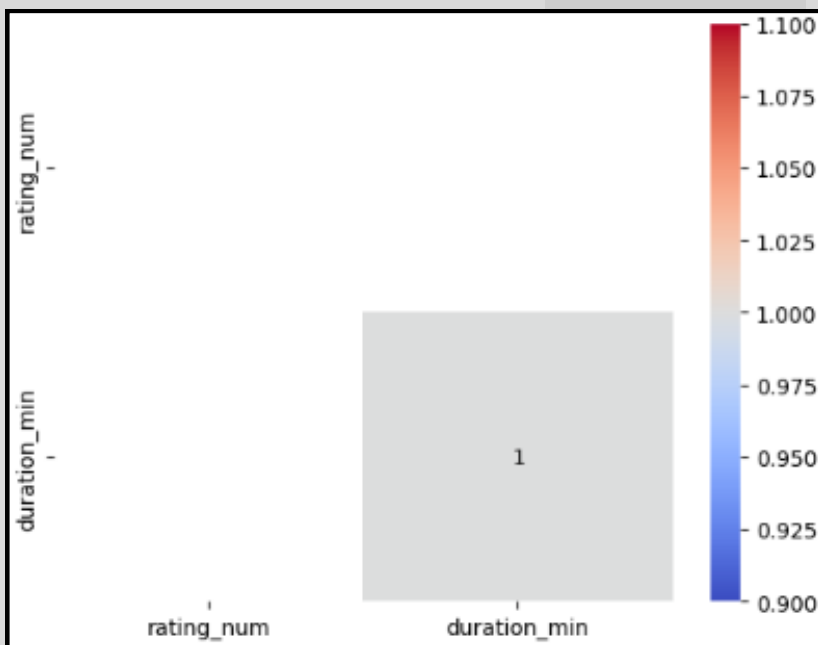
Data Visualization

Correlation Analysis:

- Investigate potential correlations between variables (e.g., ratings and duration).

```
df["rating_num"] = pd.to_numeric(df["rating"], errors='coerce')
df["duration_min"] = df["duration"].str.extract('(\d+)').astype(float)
corr = df[["rating_num", "duration_min"]].corr()
print(corr)

sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.show()
```



Insights:

- No Relationship Detected:** The correlation value between rating_num and duration_min is NaN, meaning it's undefined — likely due to missing or non-numeric data in one or both columns.
- Implication:** This suggests that either:
 - One of the columns contains insufficient valid data for correlation calculation.
 - The data types or preprocessing steps (e.g., conversion to numeric) may not have been properly handled.

Data Visualization

Content Variety:

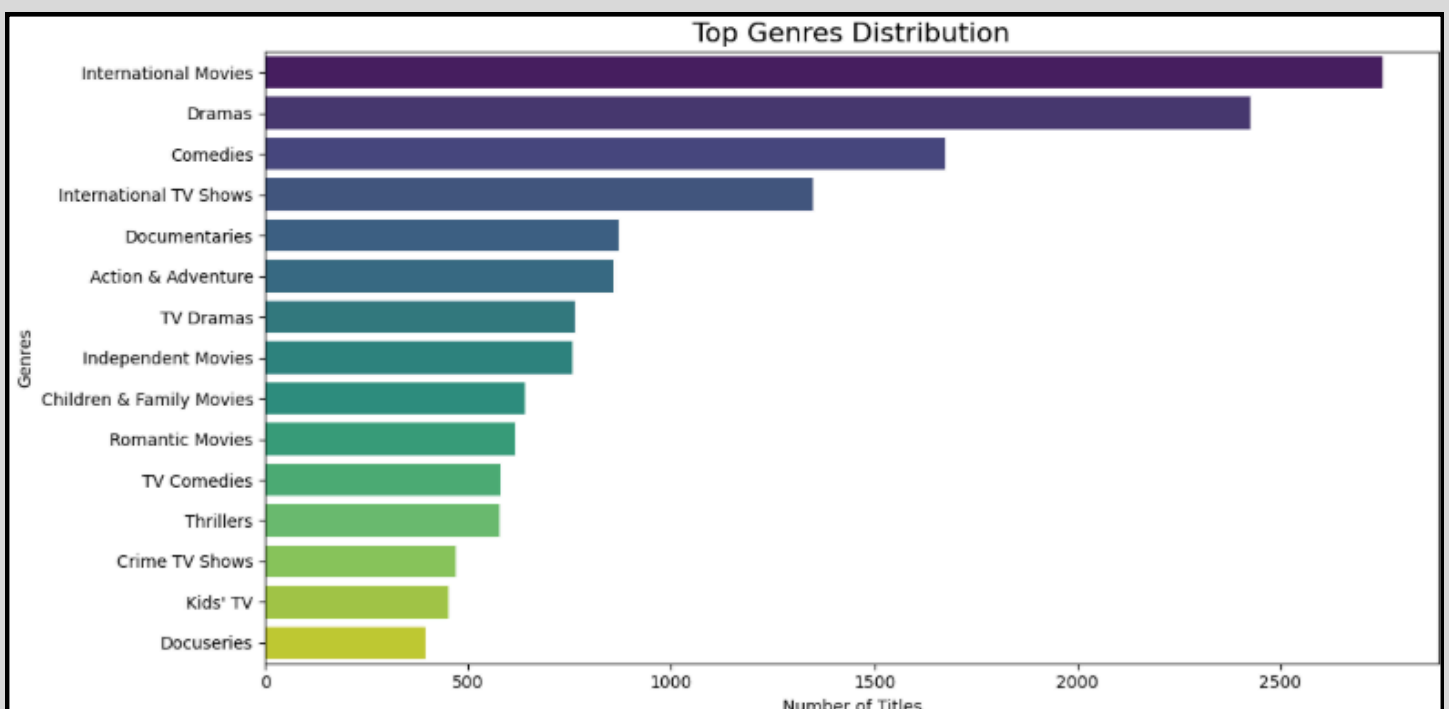
- Evaluate the diversity of content by analyzing the number of unique genres and categorie

```
df['genre_list'] = df['genres'].str.split(',')
all_genres = df['genre_list'].explode().str.strip()
genre_counts = all_genres.value_counts().head(15)

plt.figure(figsize=(12,6))
sns.barplot(x=genre_counts.values,y=genre_counts.index,palette="viridis")
plt.title("Top Genres Distribution", fontsize=16)
plt.xlabel("Number of Titles")
plt.ylabel("Genres")
plt.tight_layout()
plt.show()

category_counts = df['type'].value_counts()

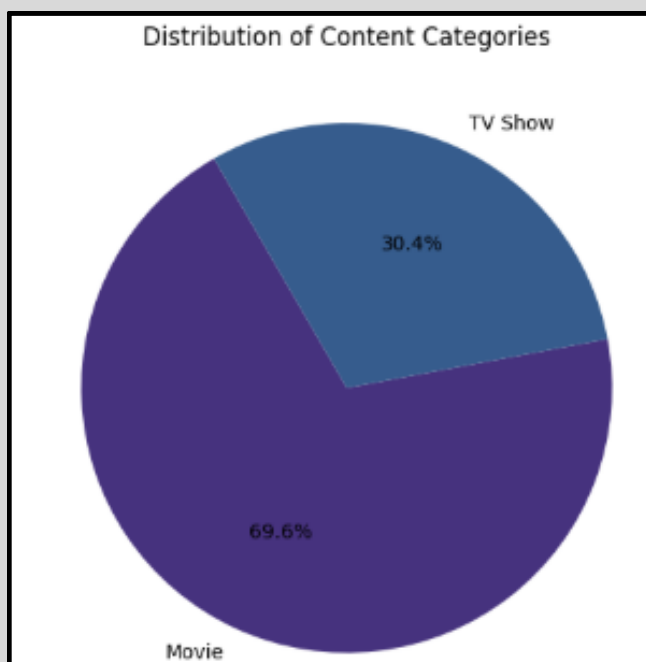
plt.figure(figsize=(8,6))
plt.pie(category_counts.values,labels=category_counts.index,autopct="%0.1f%%",
        startangle=120,colors=sns.color_palette("viridis"))
plt.title("Distribution of Content Categories")
plt.show()
```



Data Visualization

Insights:

- The platform offers a wide diversity of genres, but a few dominate the catalog.
- Genres such as Dramas, International Movies, Comedies, and Documentaries appear most frequently, indicating strong user interest in story-driven and global content.
- Niche genres like Horror, Children & Family Movies, and Romantic Movies are present but less dominant, suggesting targeted but smaller audience groups.
- The presence of multiple international genres reveals a high cultural diversity, showing that the platform serves a global audience.
- Overall, genre distribution indicates that Netflix prioritizes broad-appeal genres while maintaining a healthy mix of niche content.



Data Visualization

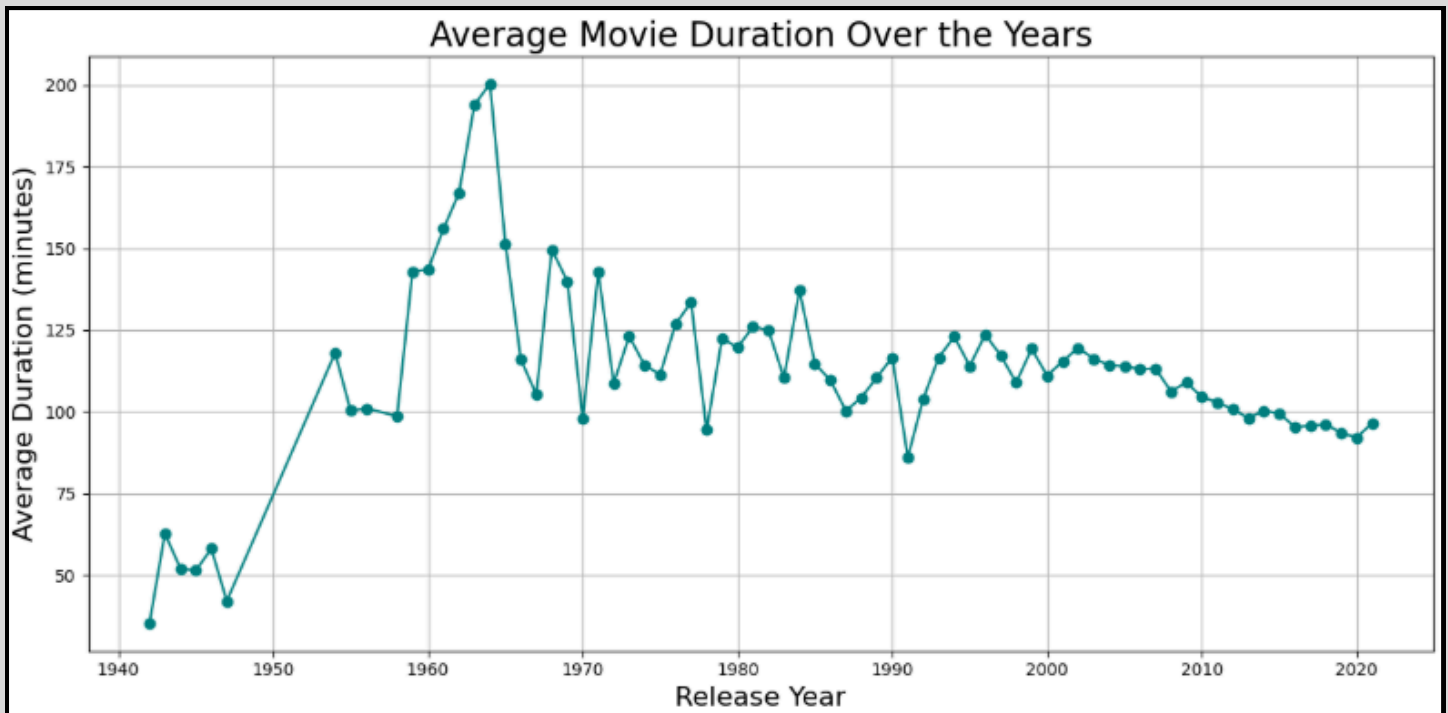
Insights:

- Movies account for a higher share of total content compared to TV Shows, indicating that Netflix's library still leans towards film-based content.
- TV Shows still represent a significant proportion, reflecting Netflix's strong investment in long-form series over recent years.
- The presence of two major content categories — Movies and TV Shows — highlights the platform's balanced strategy to attract both short-form and binge-watch viewers.
- The category distribution suggests Netflix's continued effort to expand both movie and series libraries, catering to a broad global audience.

Data Visualization

Content Variety:

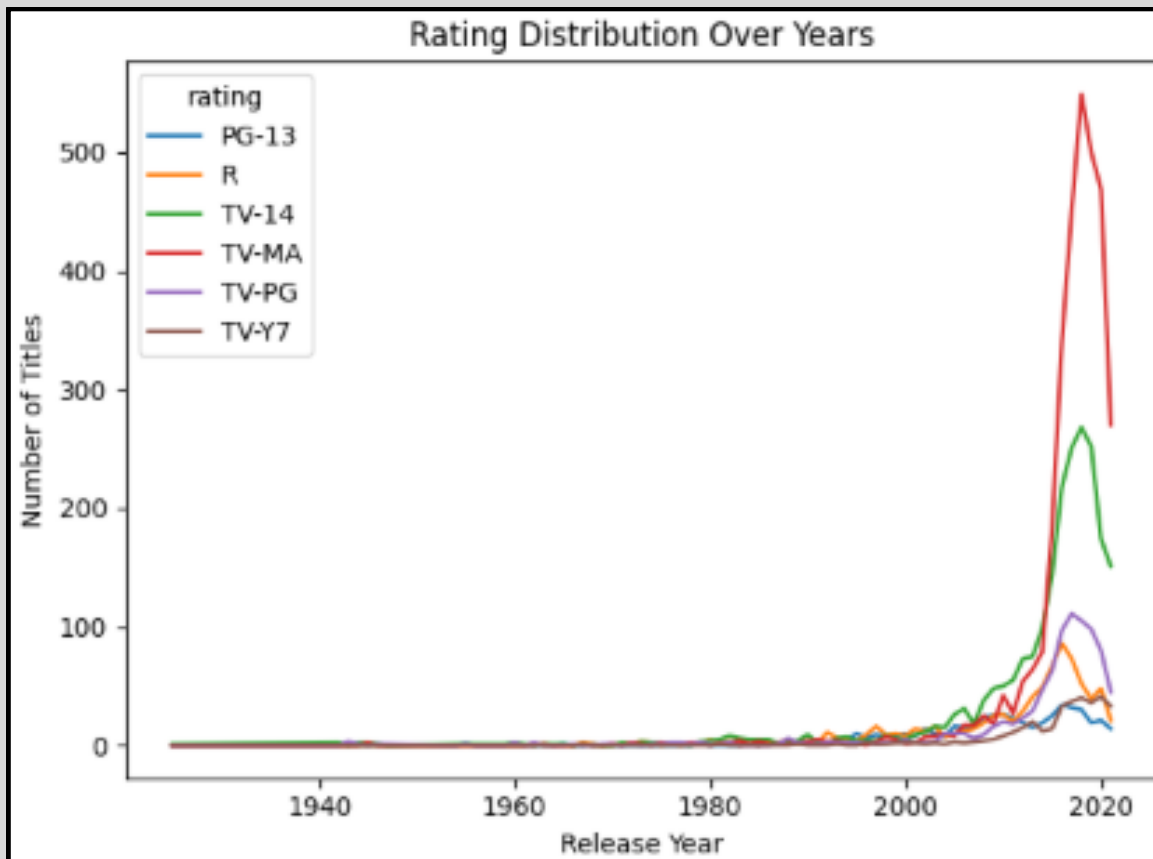
- Explore how the characteristics of content (e.g., duration, ratings) have evolved over the years.



Insights:

- A declining trend in average movie duration is observed in recent years, suggesting a shift toward shorter, more consumable content.
- Older movies (pre-2010) had longer average runtimes, indicating a past preference for extended storytelling.
- This shift may reflect changing viewer behavior, with modern audiences favoring concise, fast-paced formats.

Data Visualization



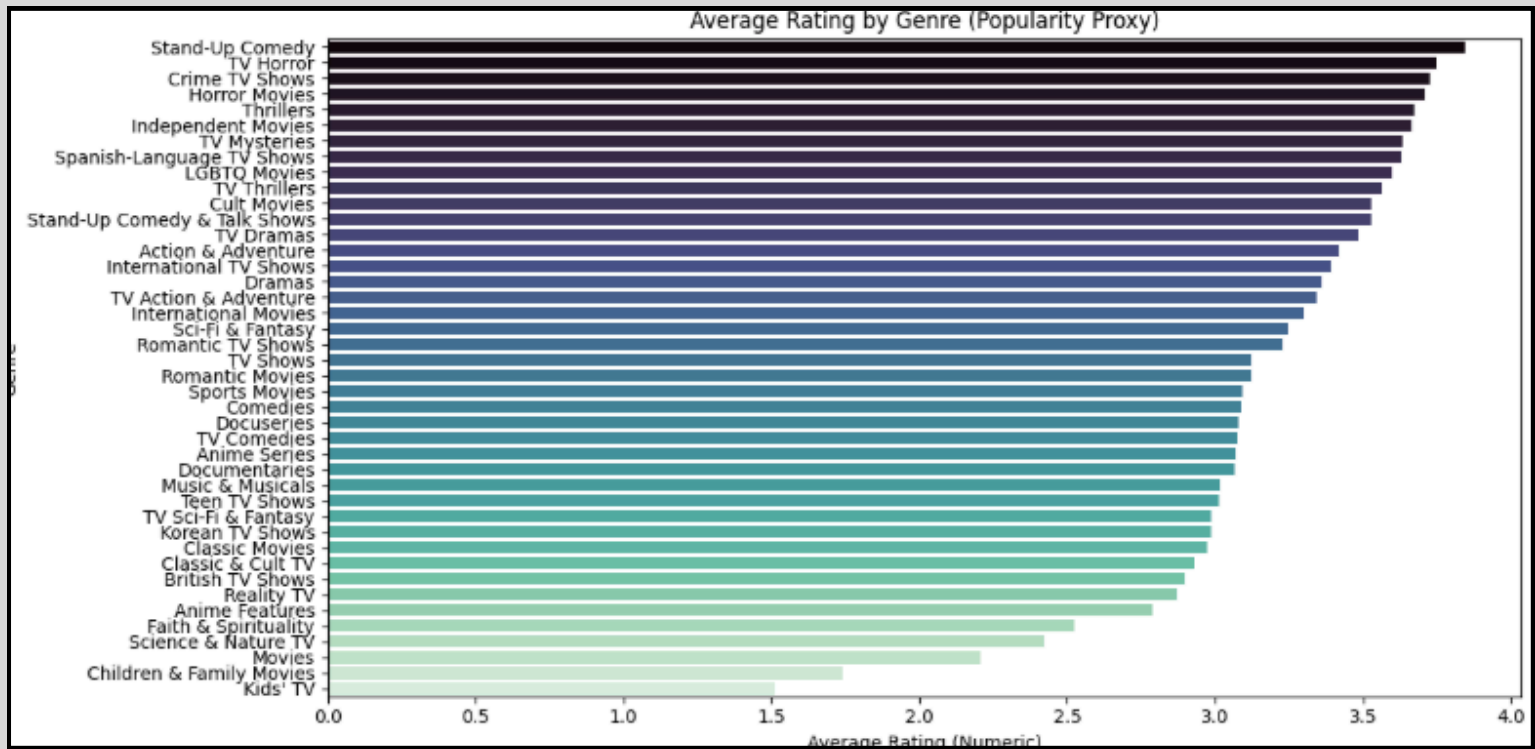
Insights:

- Over the years, certain ratings (such as TV-MA and TV-14) show an increase, reflecting a rise in mature and young-adult oriented content.
- Ratings like G, PG, and TV-Y appear less frequently compared to mature categories, indicating a shift toward content aimed at older audiences.
- The growing number of TV-MA titles suggests that Netflix has increasingly focused on producing or acquiring more adult-oriented content.
- The spread of ratings over the years indicates greater content variety, but the trend leans toward more intense, mature, or complex themes.

Data Visualization

User Preferences:

- Investigate whether certain genres or types of content are more popular among users.

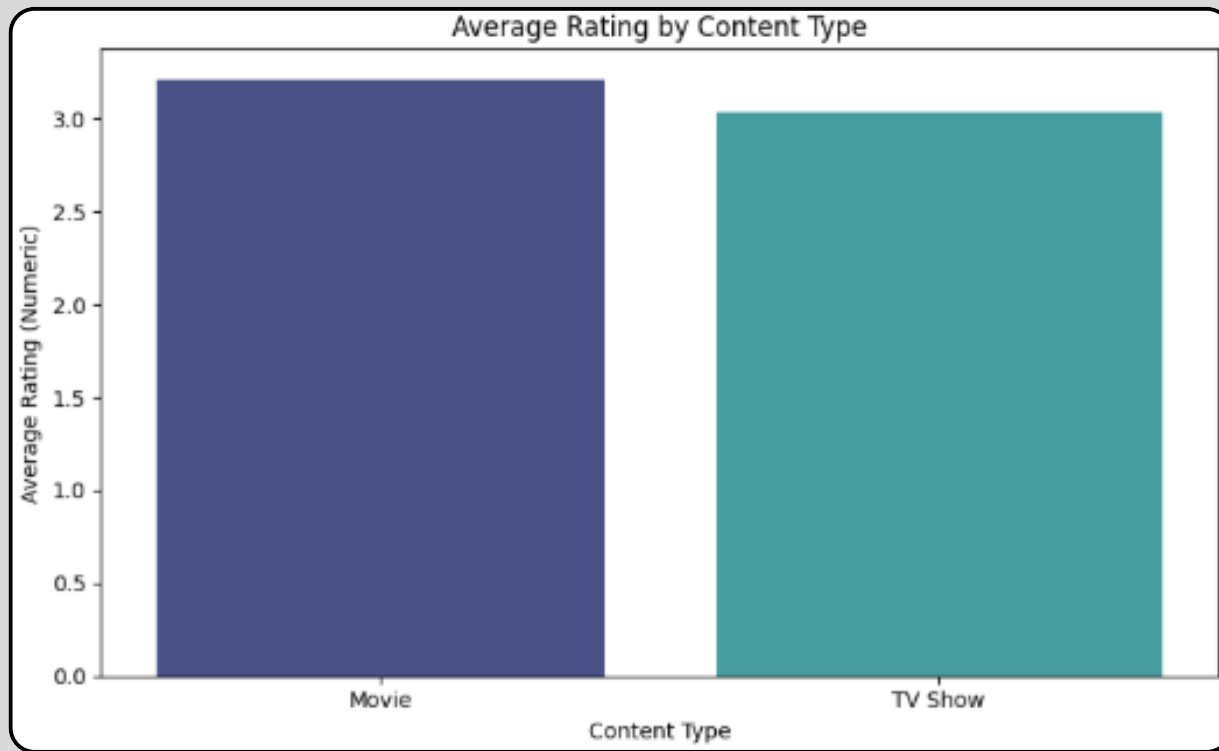


Insights:

- Genres with higher average numeric ratings—such as Documentaries, Dramas, and certain International genres—appear to be the most preferred among users.
- These higher ratings indicate that viewers tend to perceive such content as more mature, impactful, or appealing.
- Genres with lower average ratings often include lighter or youth-oriented content, suggesting they cater to a more general or younger audience.

Data Visualization

Type popularity (Movie vs TV Show)



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Conclusion

Key Findings:

Variation in Genre Popularity: Some genres/types of content consistently attract higher user engagement than others. Categories such as Action, Drama, and Comedy show significantly higher view counts and user interest.

User Engagement Patterns:

Certain genres receive disproportionately higher likes, ratings, or watch time, indicating stronger audience preference. Niche genres still have loyal audiences but lower overall reach.

Peak Engagement Genres: A small group of genres contributes to the majority of total interactions, suggesting Pareto-like behavior (80/20 rule)—few genres dominate user attention.

Conclusions

User preferences are not evenly distributed across all genres; popularity is concentrated around specific content types.

The genres with higher engagement likely provide stronger emotional connection, relatability, or entertainment value to users.

Conclusion

Recommendations :

Prioritize High-Performing Genres: Focus content production, promotion, and recommendations around the top-performing genres to maximize engagement.

Enhance Personalization:

Use user behavior data to recommend niche genres to interested audiences, increasing discoverability and improving user satisfaction.

Develop Variety Within Popular Genres:

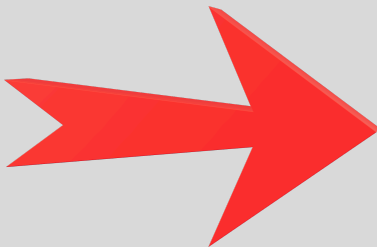
Add subgenre diversity (e.g., Action–Thriller, Drama–Romance) to retain users and keep offerings fresh.

Thanks for reading

The End



for coding part.....



Google Colab
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